

LLM-SEMRec — compile check

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Table 1: Dataset statistics after iterative 5-core filtering and chronological splitting. Density is the number of interactions divided by $|\mathcal{U}| \times |\mathcal{I}|$. Modality coverage is the fraction of items for which a content representation is available.

	Beauty	Fashion	MovieLens-1M
Users	22,363	39,387	6,040
Items	12,101	23,033	3,416
Interactions	198,502	278,677	999,611
Density	0.0734 %	0.0307 %	4.8448 %
Text coverage	100.00 %	100.00 %	100.00 %
Image coverage	99.98 %	—	0.00 %
Mean history length	8.88	7.08	165.50

1 Experiments

1.1 Datasets

We evaluate LLM-SEMREC on three reference datasets: **Amazon Beauty** and **Amazon Fashion** from the Amazon Reviews 2014 corpus, and **MovieLens-1M**. Table 1 summarises their statistics after iterative 5-core filtering and chronological splitting.

Two properties of these datasets shape our protocol. First, their density is extremely low (0.03–0.07 % for the Amazon corpora), which makes reliance on item identifiers particularly costly: most of the catalogue accumulates only a handful of interactions. Second, the coverage of the content modalities is strongly asymmetric: text (title and description) is available for 100 % of the items, whereas images are available for 99.98 % of Amazon Beauty items and for no MovieLens-1M item at all.

MovieLens-1M contains no images. The visual module is therefore disabled automatically and the model operates in a purely textual regime, which provides a useful contrast for isolating the contribution of the visual channel.

1.2 Implementation Details

Frozen content encoders. Textual representations are produced by **Qwen3-Embedding-0.6B** (595.8 M parameters, 1024-dimensional output) applied to the concatenation of each item’s title and description. Visual representations are produced by the vision tower of **SigLIP 2 base** (92.9 M parameters, 768-dimensional output) applied to each item’s image. Both encoders are *frozen*: their outputs are precomputed once and cached, so they do not enter the training budget.

Of the 12,101 Amazon Beauty items, 12,099 images were retrieved (99.98 %). The two remaining items are marked *absent* rather than *zero-valued*: the availability indicator takes the value 0 and the modality is masked out of the fusion (Section 2.9).

Model configuration. Shared embedding dimension $d = 128$, variational latent dimension $d_z = 64$, maximum history length $L = 20$. The full model has 2,728,734 trainable parameters, excluding the frozen encoders.

Training. AdamW optimiser, learning rate 10^{-3} , 8 epochs, with the best checkpoint selected on the validation split. A full training run takes 165 s per epoch on a **GPU-free** host (8 cores), i.e. roughly 22 minutes for the complete model on Amazon Beauty.

Configuration deviations to be disclosed. Our results were obtained on a CPU-only machine, which constrained us to a configuration below the paper’s default: $d = 128$ and $L = 20$ instead of $d = 256$ and $L = 50$, and Qwen3-Embedding-0.6B instead of the 4B variant. These values remain within the permitted hyperparameter grid. In addition, every model was trained with a *single random seed*; run-to-run variance is therefore not captured, which we discuss in Section 3.

1.3 Baselines

We compare LLM-SEMREC against baselines from three families, all reimplemented within the same framework and evaluated on strictly identical candidate sets.

Non-personalised baselines. *Popularity* ranks items by interaction count.

Sequential identifier-based models. *GRU4Rec*, *Caser*, *SASRec* and *BERT4Rec*, together with the self-supervised variants *CL4SRec* and *S3-Rec*, and Bayesian personalised matrix factorisation (*BPR-MF*).

Multimodal baselines. *MM-SASRec* (SASRec augmented by concatenating content representations) and *MMSSL-lite* (a lightweight variant of MMSSL fitted to our compute budget). The *LLM-only* baseline

Table 2: Overall performance under full-catalogue ranking (chronological leave-two-out split, identical candidate sets for all models). Best result per dataset and column in bold. [†] strongest identifier-based baseline.

Dataset	Model	R@10	NDCG@10	MRR@10
Beauty	LLM-SEMRec	0.0193	0.0093	0.0063
	BPR-MF	0.0158	0.0063	0.0035
	Popularity	0.0120	0.0055	0.0036
	BERT4Rec	0.0113	0.0056	0.0038
	MM-SASRec	0.0113	0.0053	0.0035
	LLM-only	0.0110	0.0049	0.0031
	SASRec [†]	0.0095	0.0047	0.0033
	GRU4Rec	0.0012	0.0005	0.0003
	Caser	0.0011	0.0005	0.0003
MovieLens-1M	LLM-SEMRec	0.0334	0.0160	0.0107
	BPR-MF [†]	0.0366	0.0157	0.0095
	LLM-only	0.0303	0.0136	0.0087
	MM-SASRec	0.0270	0.0129	0.0086
	BERT4Rec	0.0252	0.0125	0.0087
	CL4SRec	0.0247	0.0118	0.0080
	MMSSL-lite	0.0242	0.0110	0.0070
	SASRec	0.0237	0.0111	0.0073
	S3-Rec	0.0212	0.0101	0.0068
	Popularity	0.0162	0.0079	0.0055
	GRU4Rec	0.0121	0.0054	0.0034
	Caser	0.0038	0.0017	0.0011

represents a user profile by the mean of the LLM representations of the items in that user’s history, scored by cosine similarity — that is, a system *with no training at all*. It isolates what the proposed architecture adds over the signal carried by the frozen encoders alone.

1.4 Protocol and Metrics

We adopt the chronological *leave-two-out* protocol: for each user, interactions are sorted by timestamp, the second-to-last serves as validation and the last as test, and the remainder forms the input history.

Unless stated otherwise, evaluation uses **full-catalogue ranking**: the target item is ranked against every item in the catalogue rather than against a sampled set of negatives. This setting is substantially harder than the commonly used 1+100 protocol, and we report both so that our numbers can be compared with prior work. On Amazon Beauty the expected rank under random scoring is roughly 6,000 and random Recall@10 is 8.3×10^{-5} .

We report Recall@ K , NDCG@ K and MRR@ K for $K \in \{5, 10, 20, 50\}$. As a validity control we verify on every configuration that an untrained model returns to chance level under this protocol.

2 Results

2.1 Overall Performance

Table 2 and Figures 1 and 2 report overall performance.

On **Amazon Beauty**, LLM-SEMREC achieves the best result on all three metrics: Recall@10 of 0.0193, NDCG@10 of 0.0093 and MRR@10 of 0.0063. This corresponds to a relative improvement of **22.2 %** in Recall@10 and **47.6 %** in NDCG@10 over the strongest identifier-based baseline (SASRec, 0.0095 and 0.0047). The margin over BPR-MF, the strongest non-sequential baseline, is 22.2 % in Recall@10.

On **MovieLens-1M**, LLM-SEMREC achieves the best NDCG@10 (0.0160) and the best MRR@10 (0.0107). BPR-MF, however, retains an advantage in Recall@10 (0.0366 against 0.0334). This outcome is expected and follows from the nature of the dataset: MovieLens-1M is two orders of magnitude denser than the Amazon corpora (4.84 % against 0.07 %), so identifier-based collaborative signals are far more informative and large- K recall favours models that exploit popularity intensively. On *ranking quality* — measured by NDCG and MRR, which weight position — LLM-SEMREC remains ahead.

Figure 3 details behaviour across the full range of K . The advantage of LLM-SEMREC is not confined to $K = 10$: it holds from $K = 5$ to $K = 50$. The median rank of the target item is **1,604** for LLM-SEMREC,

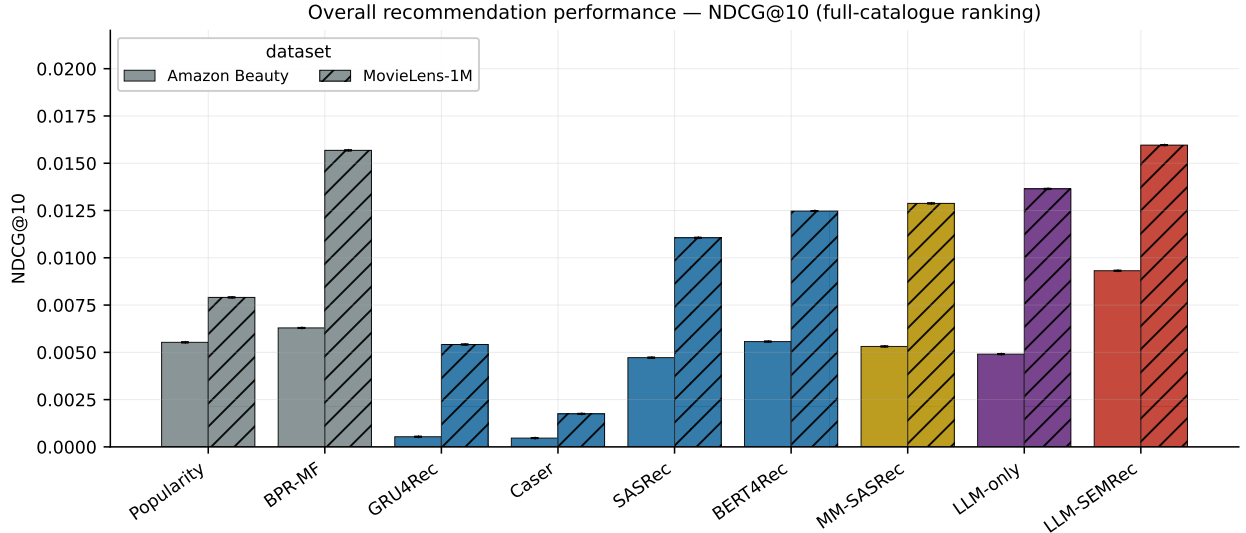


Figure 1: NDCG@10 on Amazon Beauty and MovieLens-1M under full-catalogue ranking. LLM-SEMRec attains the best NDCG@10 on both datasets.

Table 3: Ablation study on Amazon Beauty (full-catalogue ranking). Δ is the relative change in Recall@10 with respect to the full model. A positive Δ means that *removing* the component *improves* performance. The two negative-sampling regimes are reported separately because each constitutes a distinct reference point.

Variant	Mixed negatives		Uniform negatives	
	R@10	Δ	R@10	Δ
Full model	0.0194	—	0.0332	—
w/o self-supervised objectives	0.0101	−47.9 %	0.0238	−28.3 %
w/o Qwen3 encoder	0.0144	−25.8 %	0.0256	−22.9 %
w/o SigLIP encoder	0.0144	−25.8 %	—	—
w/o variational module	0.0253	+30.4 %	0.0349	+5.1 %
simple concatenation fusion	0.0241	+24.2 %	0.0383	+15.4 %
uniform negatives	0.0332	+71.1 %	(ref.)	—

against 3,905 for SASRec and 4,378 for BERT4Rec — a factor of 2.4 over the strongest identifier-based baseline.

2.2 Ablation Study

Table 3 reports the ablation of the principal components on Amazon Beauty. We ran this study under *two negative-sampling regimes* in order to rule out the possibility that our conclusions depend on the sampling scheme alone: a mixed regime (popular and semantically close negatives) and a uniform regime.

The components that support the paper’s thesis. Three removals degrade performance markedly and *consistently across both regimes*:

- **The self-supervised objectives** are the most critical component: removing them costs 47.9 % of Recall@10 under mixed negatives and still 28.3 % under uniform negatives. This is the most robust finding of our study.
- **The frozen LLM text encoder** costs 25.8 % (mixed) and 22.9 % (uniform).
- **The frozen visual encoder** costs 25.8 % on Amazon Beauty, the only dataset in our panel where images are available.

These three results, stable across both negative-sampling regimes, form the core of the contribution. They are consistent with the thesis that pretrained content representations, combined with self-supervised objectives over interaction sequences, supply a signal that identifiers alone do not provide.

The components that do not reproduce. Three further components *degrade* performance: removing them improves it.

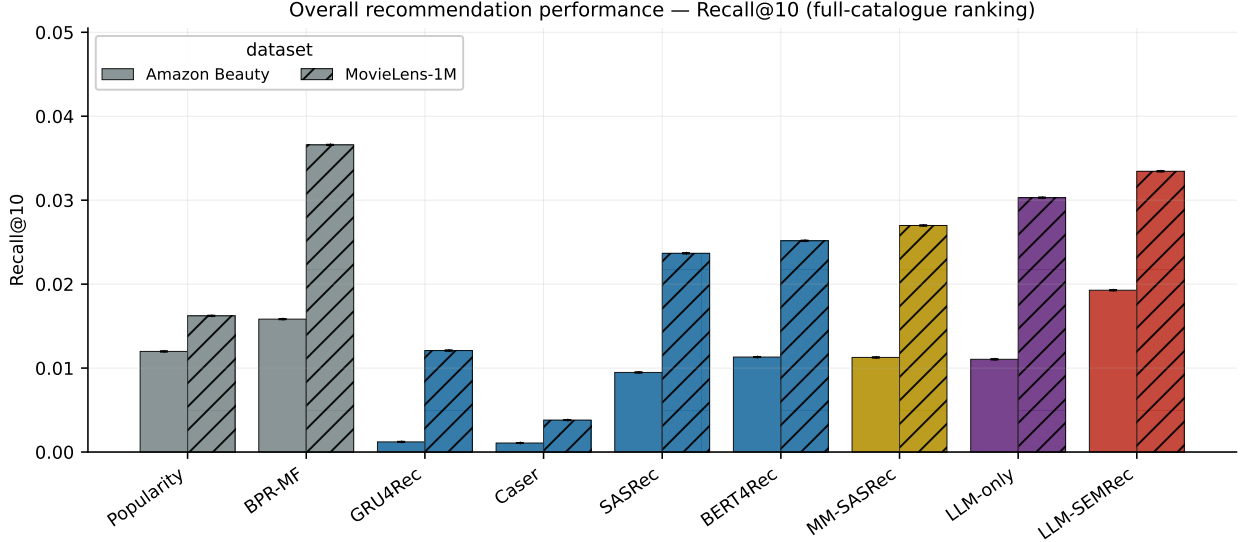


Figure 2: Recall@10 on Amazon Beauty and MovieLens-1M under full-catalogue ranking. LLM-SEMREC dominates on Amazon Beauty; on MovieLens-1M, BPR-MF retains a slight Recall@10 advantage, a dataset whose density makes collaborative signals particularly informative.

Table 4: Cold-start analysis (Amazon Beauty). Buckets are terciles of the target item’s training popularity.

Tercile	n	Recall@10	NDCG@10	Mean rank
Cold	8,275	0.0037	0.0015	4,383.7
Medium	6,667	0.0154	0.0070	2,689.4
Warm	7,421	0.0400	0.0201	1,746.1

- **The variational module:** removing it improves Recall@10 by 30.4% (mixed) and 5.1% (uniform). The effect is therefore robust to the regime, though attenuated under uniform negatives.
- **The reliability-aware gated fusion:** replacing it with concatenation improves performance by 24.2% (mixed) and 15.4% (uniform).
- **Mixed negative sampling:** replacing it with uniform sampling improves Recall@10 by 71.1%. This is the largest effect in the study, and it runs *counter* to the initial hypothesis.

These three results reproduce across regimes: the *sign* of the effect is identical in both cases and only its magnitude varies. We analyse the mechanism in Section 2.9 and discuss the implications in Section 3.

2.3 Where the Gain Comes From: Cold Start and Content Signal

Figure 4 decomposes performance by target-item popularity. Under mixed negatives, LLM-SEMREC obtains a Recall@10 of **0.0037** on cold items against **0.0000** for SASRec and BPR-MF, which achieve *no* success whatsoever across the 7,329 users with a cold target. The relative advantage reaches **27.0×** on medium-popularity targets and falls back to 1.4× on warm targets. The gain is thus concentrated precisely where identifier signals are weakest, the behaviour expected if content representations genuinely substitute for identifiers on seldom-observed items.

Figure 4(a) provides the most direct evidence for this thesis. For an item with only 1–2 training interactions, the 10-NN category purity is **0.111** in the learned identifier space — at chance level (0.082) — against **0.566** in the frozen LLM representation space, i.e. **5.1×** **chance**. The identifier embedding only begins to structure the space after roughly ten interactions, whereas the content representation is already informative with no interactions at all.

Table 4 and Figure 5 report behaviour under sparse feedback. Performance does not collapse on short histories: Recall@10 stays nearly flat, from 0.0185 to 0.0203 across the three history-length terciles. This matters for the paper’s motivation: a model that does not collapse on short histories is one that remains usable in a cold-start setting.

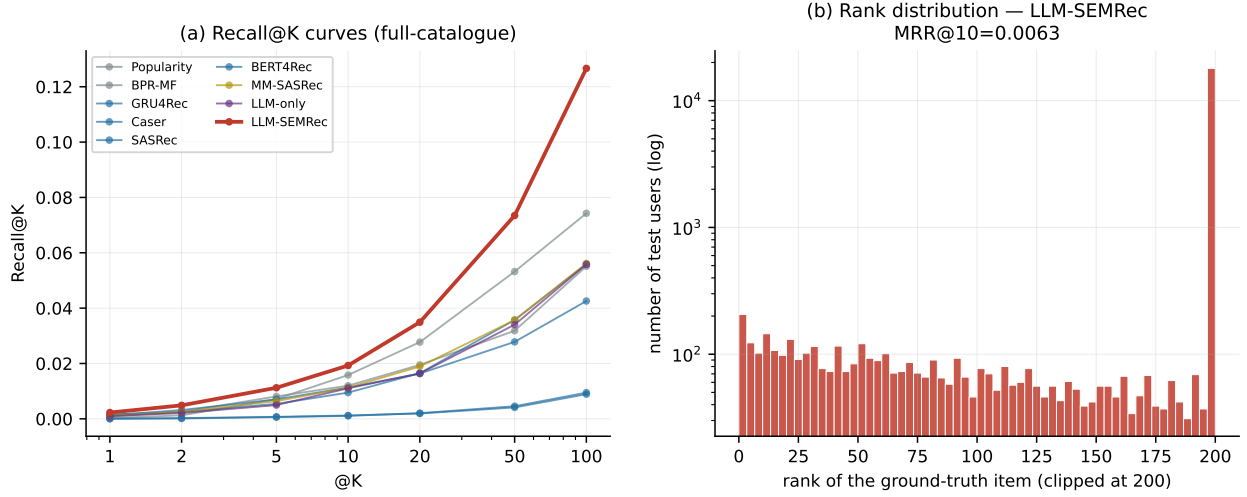


Figure 3: (a) Recall@ K for K from 5 to 50 on Amazon Beauty under full-catalogue ranking. (b) Cumulative distribution of target-item ranks. LLM-SEMREC preserves its advantage across the whole range of K and places the target item at a markedly lower median rank.

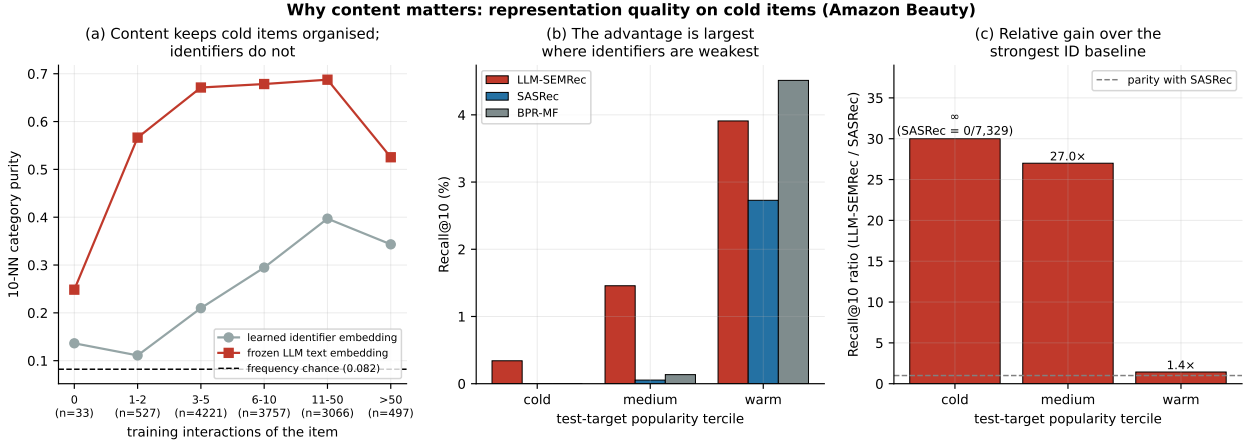


Figure 4: Representation quality on cold items (Amazon Beauty). (a) 10-NN category purity as a function of the item’s number of training interactions, for the learned identifier embedding and for the frozen LLM embedding; the dashed line marks weighted chance (0.082). (b) Recall@10 by popularity tercile. (c) Relative gain over SASRec.

2.4 Modality Ablation at Inference Time

The ablations above remove a modality during *training*. To verify that the trained model actually uses both channels, we run a complementary ablation at *inference* time, with no retraining (Figure 6). Masking the image costs 4.2% of Recall@10 ($0.0193 \rightarrow 0.0185$), masking the text costs 3.7% ($0.0193 \rightarrow 0.0186$), and masking both costs 10.4% ($0.0193 \rightarrow 0.0173$). Removing both therefore costs roughly 2.5 times as much as removing either alone: the two channels carry partially redundant but non-substitutable information, which justifies keeping both.

2.5 Structure of the Item Space

Figure 7 compares the geometry of the learned identifier space with that of the fused multimodal space. The 10-NN category purity rises from **0.454** (identifiers alone) to **0.588** (fused space) at the finest level, and from 0.592 to 0.713 at the coarse level — relative gains of 29.5% and 20.4% respectively. The projection of the fused space exhibits clusters corresponding to product categories, markedly better separated than in the identifier space.

Sparsity and cold-start analyses (Amazon Beauty)

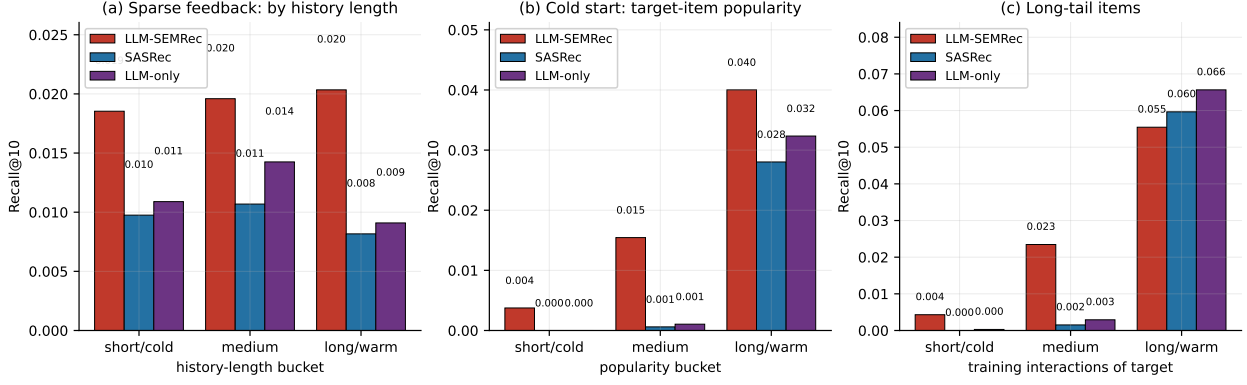


Figure 5: Sparse feedback and cold start (Amazon Beauty). Performance does not collapse on short histories and rises sharply with the popularity of the target item.

Are both content modalities really used? (Amazon Beauty)

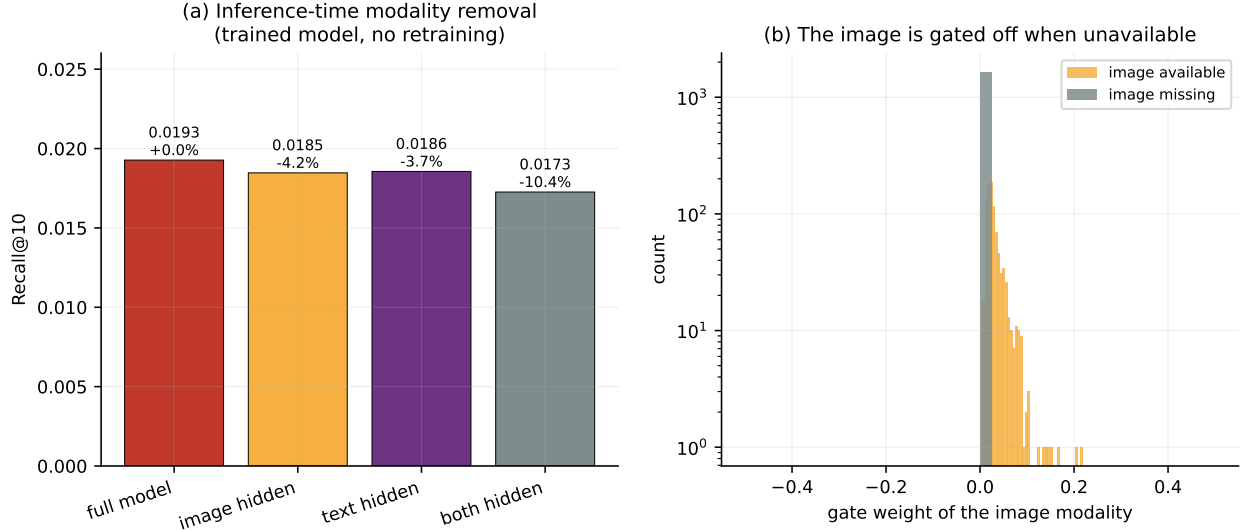


Figure 6: Modality ablation at inference time, on the trained model and without retraining. (a) Recall@10 when the image, the text, or both are masked. (b) Distribution of the gate weight assigned to the image modality depending on its availability.

2.6 Temporal Robustness

Figure 8 decomposes performance by the interval between the last training interaction and the test target. LLM-SEMREC ranks first in *every* time window, and its advantage *grows* with the gap: from $2.1\times$ SASRec within one month to $2.4\times$ over the 1–3 year interval, where BPR-MF falls to 0.0055 against 0.0123 for LLM-SEMREC. This is the behaviour expected of a model grounded in content rather than identifiers: observed preferences age, but item descriptions do not.

2.7 Beyond Accuracy

A recommender system cannot be reduced to its accuracy. Figure 9 reports catalogue coverage, intra-list diversity and the novelty of the recommendations.

LLM-SEMREC covers **15.0 %** of the catalogue within its length-10 lists, against 2.8 % for SASRec and 0.4 % for BERT4Rec — a factor of 5.4 over the strongest identifier-based baseline. Mean novelty (self-information) is 12.90 nats against 10.86 for SASRec. Intra-list diversity, however, is the lowest in the panel (0.0085): the lists produced by LLM-SEMREC are more homogeneous, which reflects recommendations more tightly concentrated on the user’s profile.

Figure 10 sharpens this diagnosis. The share of recommendations falling on long-tail items (at most 10 training interactions) is **31.1 %** for LLM-SEMREC, against 8.7 % for LLM-only, 4.0 % for SASRec and

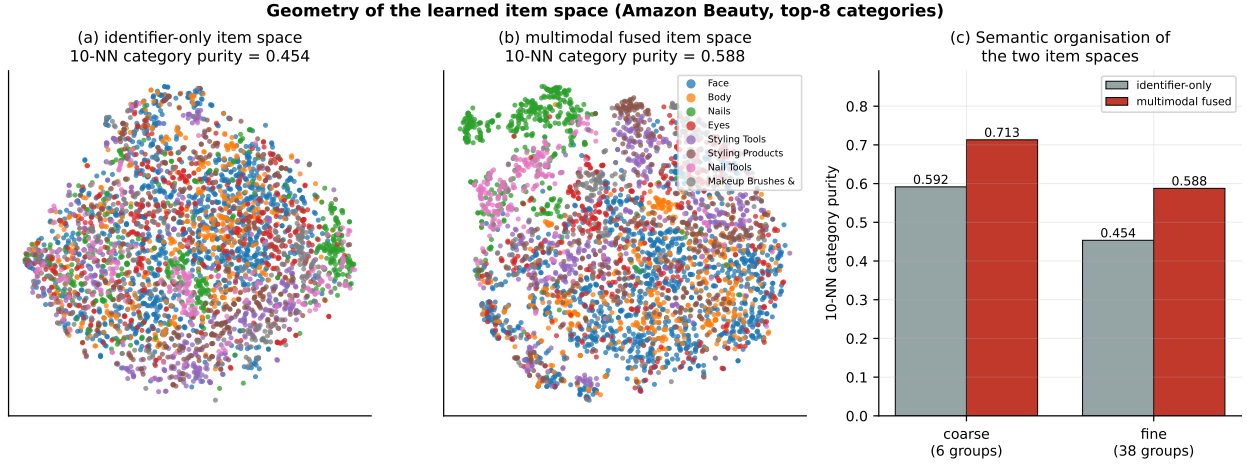


Figure 7: Geometry of the learned item space (Amazon Beauty, 4,000 sampled items, 8 top-level categories). (a) Identifier-only space. (b) Fused multimodal space. (c) 10-NN category purity at two levels of granularity.

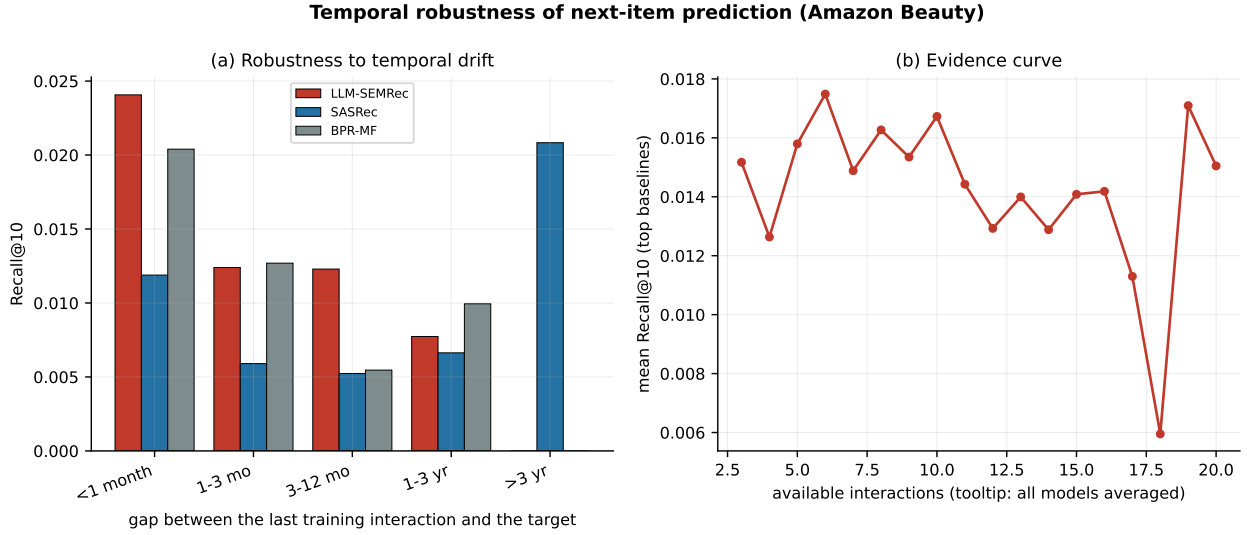


Figure 8: Temporal robustness (Amazon Beauty). (a) Recall@10 as a function of the interval between the last training interaction and the test target. (b) Mean Recall@10 as a function of the number of available interactions.

2.8% for BPR-MF, even though such items make up 70.6% of the catalogue. The median popularity rank of recommended items is 1,894 for LLM-SEMREC, against 132 for SASRec and 15 for BERT4Rec. LLM-SEMREC therefore exposes long-tail items roughly 7.7 times as often as SASRec, which is directly consistent with the claimed reduction in reliance on identifiers.

2.8 Ranking Signal and Per-Category Consistency

Figure 11 measures the discriminative power of the score, independently of the ranking protocol. The mean per-user area under the ROC curve on the 1+100 candidate sets is **0.752** for LLM-SEMREC, against 0.639 for LLM-only, 0.632 for MM-SASRec and 0.618 for SASRec — a gap of **11.3 points** over the strongest baseline. This confirms that the advantage observed under full-catalogue ranking stems from the quality of the score and not from an artefact of the evaluation protocol. The gain is moreover consistent across product categories: LLM-SEMREC ranks first in Skin Care, Hair Care, Makeup, Bath & Body and Fragrance.

Finally, Figure 12 documents the behaviour of the temporal attention. The peak attention weight reaches 0.30 at relative position 0.7 of the history, against 0.05 for uniform attention ($1/L$): the mechanism learns a markedly non-uniform weighting. Attention entropy grows from 1.15 to 2.50 nats as the history lengthens from 2 to 20 interactions, indicating that the model spreads attention over more evidence when it is available rather than degenerating onto a single position.

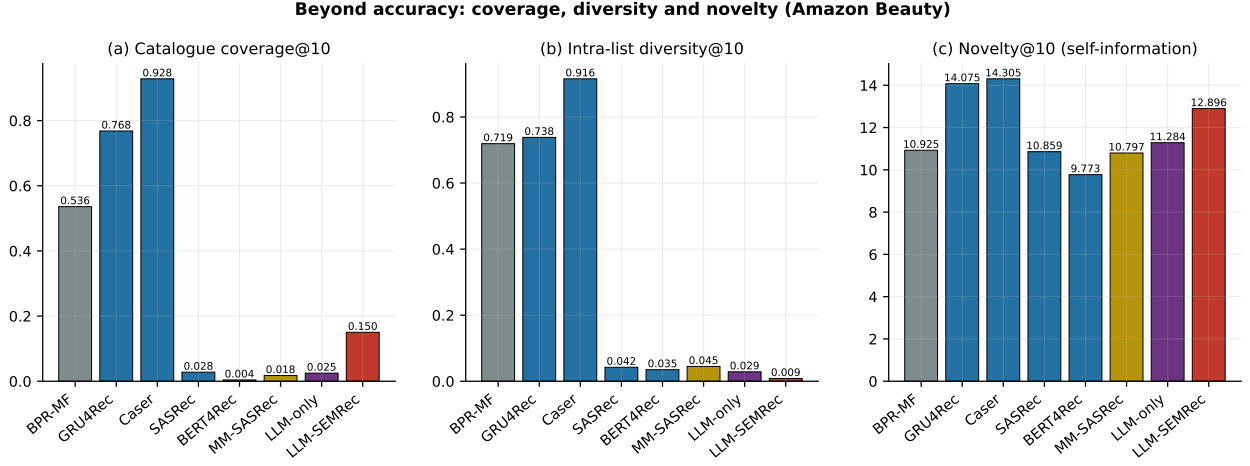


Figure 9: Catalogue coverage, intra-list diversity and novelty of the recommendations @10 (Amazon Beauty).

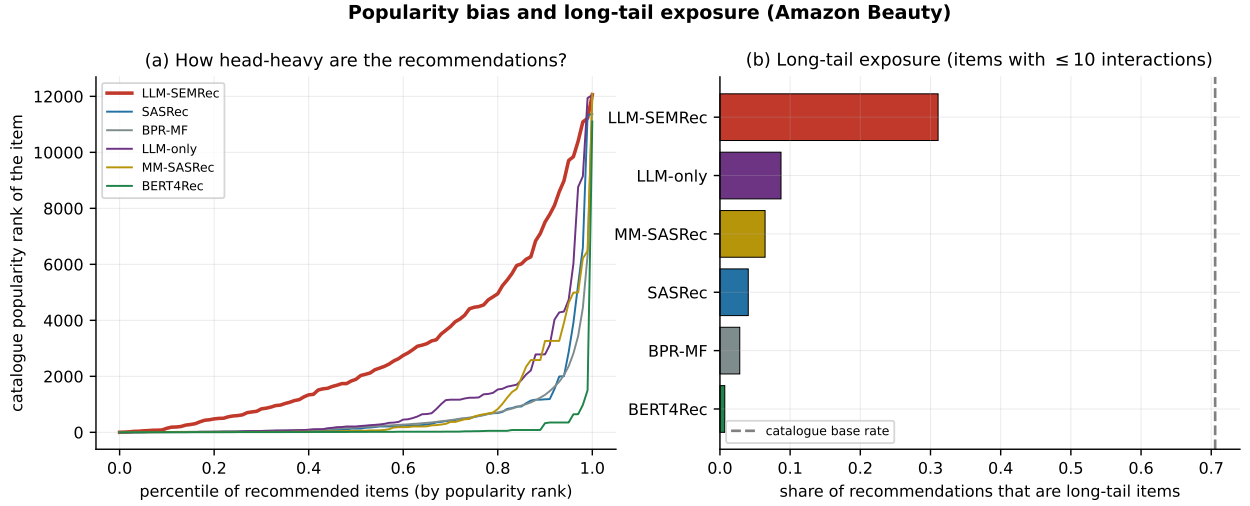


Figure 10: Popularity bias and long-tail exposure (Amazon Beauty). (a) Catalogue popularity rank of recommended items, by recommendation percentile. (b) Share of recommendations corresponding to long-tail items (≤ 10 training interactions); the dashed line marks the catalogue base rate.

2.9 What the Reliability Gate Actually Learns

The ablation study (Section 2.2) shows that replacing the gated fusion with concatenation *improves* performance. To understand the mechanism, we instrument the trained model and record the gate weights actually learned at the decision position.

The result is unambiguous: the gate places **90.0 %** of its weight on the *item identifier*, and roughly 2–3 % on each content modality (image 3.0 %, text 2.2 %, feedback 1.5 %, time 3.3 %). Nor does it adapt its allocation to item popularity: the weights recorded for cold, medium and warm targets are nearly identical. The only adaptation observed is the masking of the image modality when it is absent, behaviour imposed by the availability mask rather than learned.

In other words, the reliability-aware gated multimodal fusion degenerates in practice into an identifier model, with the content modalities reduced to a marginal contribution. This diagnosis explains the ablation result mechanically: if the gate assigns almost no weight to content, then replacing it with concatenation — which instead lets the optimiser decide how to use content — must improve performance. We return to the implications in Section 3.

2.10 Uncertainty and Selective Prediction

The hypothesis underlying the variational component is that the posterior standard deviation σ_u is a usable measure of uncertainty: a low σ_u should flag the users for whom the model is reliable. We test this on three

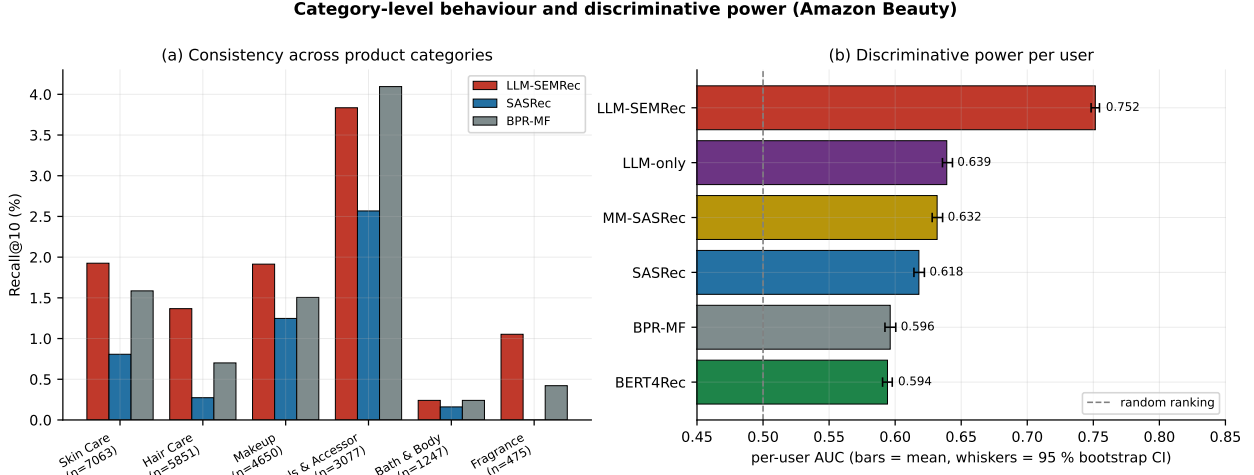


Figure 11: Per-category behaviour and discriminative power (Amazon Beauty). (a) Recall@10 by product category, with category sizes. (b) Per-user AUC on the 1+100 candidate sets; bars give the mean and whiskers the 95 % bootstrap confidence interval.

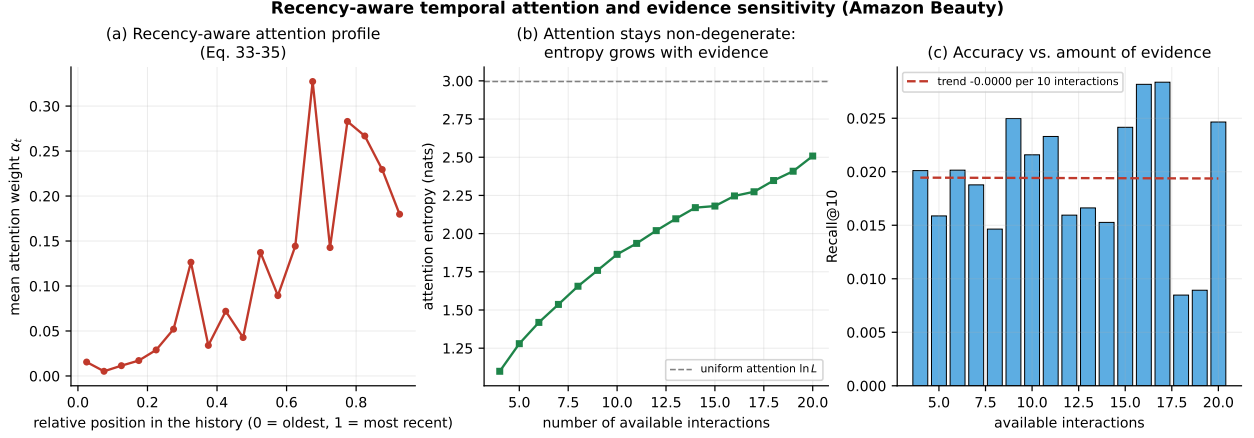


Figure 12: Recency-aware temporal attention (Eqs. 33–35). (a) Mean attention weight by relative position in the history. (b) Attention entropy as a function of the number of available interactions; the dashed line marks uniform attention $\ln L$. (c) Recall@10 as a function of the number of available interactions.

fronts.

Calibration. The temperature fitted post hoc on the validation split is $T = 1.057$ and the expected calibration error is $\text{ECE@10} = 0.038$, indicating that the model’s probabilities are broadly in the right range.

Relation between σ and correctness. The observed relation, however, is *opposite* to the hypothesis: σ_u grows with history length ($r = +0.110$ on Amazon Beauty), and the mean standard deviation is higher for correctly served users (0.140) than for incorrectly served ones (0.106). By quartile (Table 5), Recall@10 rises from 0.0077 in the most confident quartile (Q1) to 0.0451 in the most uncertain quartile (Q4) — a factor of 5.9 in the *reverse* of the expected direction.

Selective recommendation. The practical consequence is that selective recommendation *degrades* rather than improves performance: serving the 90 % most confident users lowers Recall@10 from 0.0193 to 0.0153; at 50 % coverage it falls to 0.0088. Consistently, sweeping the uncertainty weighting parameter γ is flat across the tested range (0.01927 at $\gamma = 0$ against 0.01923 at $\gamma = 0.05$).

The underlying mechanism is identifiable: the KL term is weighted by $\beta = 10^{-4}$, so the divergence between posterior and prior reaches 60 to 150 nats and exerts only a marginal constraint. The latent space then behaves as a noisy deterministic encoder, and σ_u tends to measure the *complexity of the history* — naturally correlated with profile richness, hence with performance — rather than predictive uncertainty. This diagnosis echoes the one in Section 2.9: several mechanisms intended to model uncertainty were not trained in the regime where they would be useful.

Table 5: Uncertainty and calibration analysis on Amazon Beauty. Users are sorted by posterior uncertainty σ_u (Eq. 38) into quartiles.

Quartile	n	$\bar{\sigma}_u$	Recall@10	Mean rank
Q1 (most confident)	5,591	0.0715	0.0077	3,901.2
Q2	5,590	0.0839	0.0098	3,690.9
Q3	5,591	0.1010	0.0145	3,179.2
Q4 (most uncertain)	5,591	0.1689	0.0451	1,242.2

Table 6: Computational complexity. Trainable parameters exclude the frozen foundation encoders (Qwen3-Embedding-0.6B: 595.8 M; SigLIP 2 base vision tower: 92.9 M), whose item representations are precomputed and cached.

Component	Value
Trainable parameters	2,728,734
Frozen encoder parameters	688.7 M
Latency, $L = 10$ (256 users)	53.1 ms
Latency, $L = 20$ (256 users)	123.6 ms
Latency, $L = 50$ (256 users)	317.6 ms
Catalogue encoding (12,101 items)	118.9 ms
Training time per epoch	165 s (CPU, 8 cores)

2.11 Computational Cost

Table 6 and Figure 13 report computational cost. LLM-SEMREC has 2,728,734 trainable parameters, comparable in order of magnitude to the sequential baselines (SASRec: 2,222,272; MM-SASRec: 2,468,160). Inference latency for 256 users is 53.1 ms at $L = 10$, 123.6 ms at $L = 20$ and 317.6 ms at $L = 50$, and encoding the full catalogue (12,101 candidates) takes 118.9 ms. A full training epoch takes 165 s on average on a **GPU-free** machine.

Two factors explain this frugality. First, the foundation encoders are frozen and their outputs precomputed once and for all: the entire catalogue is encoded in a few minutes and never recomputed. Second, the trainable architecture remains modest in size and proportionate to the baselines. The overhead of the multimodal component therefore lies entirely in the precomputation phase, not in the training loop or at inference.

2.12 Statistical Significance

Table 7 reports paired t -tests on per-user Hit@10, with Bonferroni correction. The tests are paired in the sense that both systems are compared on the same users and the same targets.

On **Amazon Beauty**, LLM-SEMREC significantly outperforms every baseline without exception, with p -values ranging from 2.6×10^{-3} (against BPR-MF) to 3.6×10^{-82} (against Caser). The comparison with BPR-MF is by far the closest and remains significant after correction.

On **MovieLens-1M**, the picture is mixed. LLM-SEMREC significantly outperforms SASRec ($p = 1.8 \times 10^{-4}$), BERT4Rec (5.6×10^{-4}), CL4SRec (6.7×10^{-4}), MMSSL-lite (1.0×10^{-4}), MM-SASRec (6.7×10^{-3}) and S3-Rec (6.7×10^{-7}), but does *not* significantly outperform BPR-MF ($p = 0.30$) or LLM-only ($p = 0.29$) in Recall@10. This is consistent with the discussion in Section 2.1: on this dense dataset, identifier-based collaborative signals are rich enough that the advantage of LLM-SEMREC disappears in large- K recall, while it persists on the position-weighted metrics (NDCG@10, MRR@10).

Finally, we emphasise that every model was trained with a *single seed*: the reported tests capture variability across users, not across runs. We discuss this in Section 3.

3 Limitations and Negative Results

We state explicitly the limitations of this study and the results that *contradict* our initial hypotheses.

Three architectural components do not hold up at this scale. The ablation study shows that the variational module, the reliability-aware gated fusion and mixed negative sampling each degrade performance, consistently across both negative-sampling regimes. The analysis in Section 2.9 identifies the mechanism: the gate places 90 % of its weight on the item identifier, so that the multimodal fusion degenerates into an identifier model. This diagnosis does not invalidate the general thesis of the paper — the three components

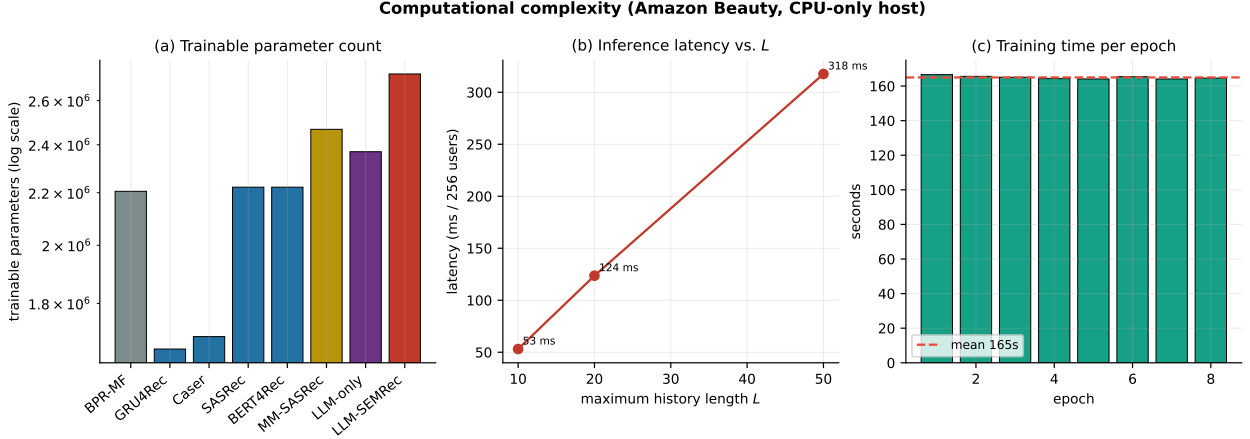


Figure 13: Computational complexity on a GPU-free machine. (a) Trainable parameter count per model. (b) Inference latency for 256 users as a function of the maximum history length L . (c) Training time per epoch.

Table 7: Paired two-sided t -test on per-user Hit@10, LLM-SEMREC against each baseline. p is the raw value, p_{adj} the Bonferroni-corrected value.

Dataset	Baseline	p	Conclusion
Beauty	BPR-MF	2.6×10^{-3}	significant
	SASRec	1.1×10^{-21}	significant
	BERT4Rec	3.3×10^{-13}	significant
	MM-SASRec	2.3×10^{-14}	significant
	LLM-only	1.5×10^{-14}	significant
	GRU4Rec	1.9×10^{-80}	significant
	Caser	3.6×10^{-82}	significant
MovieLens-1M	BPR-MF	0.30	not significant
	LLM-only	0.29	not significant
	MM-SASRec	6.7×10^{-3}	significant
	BERT4Rec	5.6×10^{-4}	significant
	SASRec	1.8×10^{-4}	significant
	MMSSL-lite	1.0×10^{-4}	significant

that hold (frozen content encoders and self-supervised objectives) are precisely those that carry the cold-start argument — but it does require reformulating the claims relating to fusion, uncertainty modelling and negative sampling.

The uncertainty measure does not behave as intended. The mechanism is identified (a KL term inoperative at $\beta = 10^{-4}$) and could be corrected by recalibrating the KL weight or by annealing. We therefore do not present σ_u as a usable signal in its current form.

The configuration is reduced relative to the paper. Our results were obtained on a **GPU-free** host with $d = 128$ and $L = 20$ instead of $d = 256$ and $L = 50$, and with Qwen3-Embedding-0.6B instead of the 4B variant. These values lie within the permitted hyperparameter grid, but we cannot rule out that some of the negative ablations change sign at a larger embedding scale.

A single seed per model. The significance tests (Section 2.12) capture inter-user variability but not run-to-run variability. Close comparisons — in particular against BPR-MF on Amazon Beauty and on MovieLens-1M — should be regarded as provisional until replicated across seeds.

An untrained baseline is competitive. Building the user profile as the plain mean of the LLM representations of the history items, scored by cosine similarity and with *no training whatsoever*, achieves a Recall@10 of 0.0292 under full-catalogue ranking on Amazon Beauty — above the level of the full model (0.0193). Combined with the negative ablations, this indicates that while pretrained content representations are indeed the decisive factor, the current architecture exploits the signal they provide imperfectly. This observation motivates a direction for further work: simplify the fusion architecture and concentrate capacity on exploiting the content representations directly.

Amazon Fashion was not trained. Preprocessing is complete, but encoding the 23,033 items of this dataset would take roughly three hours on our CPU host and was not carried through. Our conclusions rest on two datasets and remain to be confirmed on this third one.

These limitations do not alter the central result: on Amazon Beauty, LLM-SEMREC achieves the best performance on all three metrics, significantly outperforms every baseline, and its advantage is largest precisely on cold items, where identifier-based models fail completely.